

PS2 AI tutoring session (pilot)

- Deadline and credit: see the Brightspace assignment.
- Eligible problems: PS2 problems 5 (wave plates), 6 (co-propagating interference), 7 (standing waves), 8 (dipole antenna). Pick one.

What this is

You will work one PS2 problem with an AI tutor that has been set up by the instructor, and then report on the session. The tutor has been told how to behave: one question at a time, hints before formulas, no solution until you ask twice. It has also been given the instructor's notes on where each problem goes wrong. The point of the exercise is not the answer. It is to practice steering an AI toward understanding you can defend, and to notice when it is wrong.

This is a pilot for a larger program next spring. Your report is the data.

The tutor

- Sign in to Microsoft 365 Copilot with your CUNY account (via the CUNY Microsoft 365 page), then open the agent **PHY240 PS2**: [chat link](#) .
- Record the model name Copilot reports, if it reports one. If not, write "not reported".
- Do not paste personal information into the chat. Copilot is a commercial tool; the session is not private in the way office hours are.
- The tutor can be wrong. Step 5 below exists to catch that.
- If you ask for the answer repeatedly, the tutor will eventually give it. That defeats the exercise and your report will show it.
- Copilot shows clickable suggested replies under each answer. They are generated by Copilot, not by the tutor, and sometimes contain the answer. Ignore them; type your own replies. If you clicked one, say so in the report.

Protocol: one session, one problem

Do these in order. Quote your own prompts in the report, so write them with that in mind.

1. **Start.** Tell the tutor which problem. When it asks, paste the full problem statement: copy your problem's block from [phy240_ps2_problems_5-8.txt](#) (plain text, no login; identical to the problem set with the Oct 5 errata; the "Raw" button gives the cleanest copy). Then paste your own attempt, however partial (a photo of handwritten work is fine if the tutor accepts it; otherwise type what you have), and say where you got stuck.
2. **Redirect.** Ask: "Tell me which step in my attempt is weakest. Don't fix it." If the tutor starts solving, stop it and say so.

3. **Work the gap.** Proceed with hints only. Each time the tutor offers a hint, say in the chat whether you accept or reject it and why.
4. **Check one claim.** Pick one specific thing the tutor asserted. Verify it against the lecture notes, a textbook, or a person. Report the result in the chat.
5. **Compare.** Ask: "Give your full solution; list where it differs from mine." For each difference, decide who is right and say so in the chat.
6. **Variant.** Ask the tutor for a variant of the problem (one parameter or one geometric element changed). Attempt it. Decide whether it was a fair test of what you learned.
7. **Report.** Fill in `ai_session_report_template.md` and submit it as `ai_session_report_PS2-<problem number>.md` .

What is graded

Pass/fail on completeness: every field of the report filled, every quoted excerpt verbatim from your chat. Wrong physics in your attempt does not fail the report. A report with no rejected hint, no checked claim, or no identified tutor error will be read carefully; those are the fields that show you were steering.

Known limitations

- The tutor may reason badly or state a false intermediate result. Step 5 is where you catch it.
- Its restraint degrades if you push for the answer. Your excerpts should include any such moment.
- Problem 8 uses a Fourier-space method that differs from what most textbooks do; the tutor has been told to stay with the problem's method. If it drifts to "retarded potentials" or a far-field formula, redirect it and quote the exchange.